ProbOS — Month 3, Week 9
Extend REST API Model Registry

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Week 9 goal

Extend `python/server/main.py`'s model registry beyond `BatteryModel2Cell` to the three Week 4 cross-discipline models (`OptionPricerModel`, `EDQueueModel`, `ClinicalTrialModel`) – closing one of the open SBIR Phase I asks (`docs/sbir_phase1_onepager.md`: "Extend the REST API's model registry beyond `BatteryModel2Cell` to the finance/operations/clinical models already built"). This is deliberately low-risk: the endpoint pattern is already proven for one model, and the three target models are unchanged, already-validated Week 4 code.

Day-by-day plan

Day 1 – Registry design + schema audit

- Review each of the three models' constructors and prior-builder functions to determine what request-schema fields each needs (e.g. `OptionPricerModel` needs S0/K/T/r; `EDQueueModel` needs Q0; `ClinicalTrialModel` needs randomisation ratio)
- Decide whether `SimulateRequest` gets model-specific optional fields, or whether each model gets its own request schema class – lean toward the latter for type safety, following the existing `schemas.py` pattern
- Update `_build_model_and_priors()`'s registry dict to include all four models

Day 2 – /simulate for the three new models

- Extend `POST /simulate` to accept `model_name` values `"option_pricer"`, `"ed_queue"`, `"clinical_trial"`
- Verify each returns percentiles/convergence matching a direct Python `MonteCarloEngine` call, at the same `rtol=1e-10` standard already established for `BatteryModel2Cell`

Day 3 – /sensitivity for the three new models

- Extend `POST /sensitivity` similarly
- Confirm `SobolSensitivity` runs correctly against each model's own `param_dim`/`state_dim` (these differ across the four models – e.g. `ClinicalTrialModel` has `state_dim=4`, not 8 like `BatteryModel2Cell`)

Day 4 – /filter for the three new models

- Extend `POST /filter` to accept the three new models
- Confirm the existing Gaussian-observation-noise likelihood pattern is appropriate for each, or document where it is not (e.g. `ClinicalTrialModel`'s Week 8 filtering needed a fundamentally different, deterministic-state design – the `/filter` endpoint may need to expose that pattern too, or explicitly scope it out for this model with a clear error message rather than silently returning a nonsensical result)

Day 5 – Integration tests

- Extend `python/tests/test_server.py` with the same numerical-match discipline already used for `BatteryModel2Cell` (HTTP response vs direct kernel call at `rtol=1e-10`), for all three new models across all three endpoints where applicable

Day 6 – pre_week_audit.sh + CI

- Run `pre_week_audit.sh` before considering the week complete
- No new CI steps expected (existing `test_server.py` execution already covers this), but verify explicitly

Day 7 – Documentation + retrospective

- Update `docs/sbir_phase1_onepager.md`: move this item from the SBIR Ask section to a demonstrated-results bullet, since it will now be complete
- Update `README.md`'s REST API section to reflect all four registered models
- Retrospective appended to this document once complete

Exit criteria

All three POST endpoints (`/simulate`, `/sensitivity`, `/filter`, where applicable) work correctly for `OptionPricerModel`, `EDQueueModel`, and `ClinicalTrialModel`, each verified against a direct Python kernel call at `rtol=1e-10` – the same standard already established for `BatteryModel2Cell` in Week 7.